

GATE 2009, EEE Question Number 13

Abstract

This project demonstrates the behavior of the NAND gate using a Raspberry Pi Pico2W, push buttons as logic inputs, and an LED as output indicator. It validates the concept that NAND is a universal gate, from which all other logic gates can be built.

1. Components

Component	Qty
Raspberry Pi Pico2W	1
Push Buttons	2
LED	1
220Ω Resistors	3
Breadboard	1
Jumper Wires	10
Laptop with Thonny IDE	1

Table: Components used

2. Setup Instructions

- Connect button A to GPIO14 with pull-down configuration.
- Connect button B to GPIO15 similarly.
- Connect the LED anode to GPIO13 through a 220Ω resistor.
- Common ground is shared between both push buttons and the LED cathode.
- Use Thonny IDE to upload the NAND logic implementation.

3. NAND Gate Logic

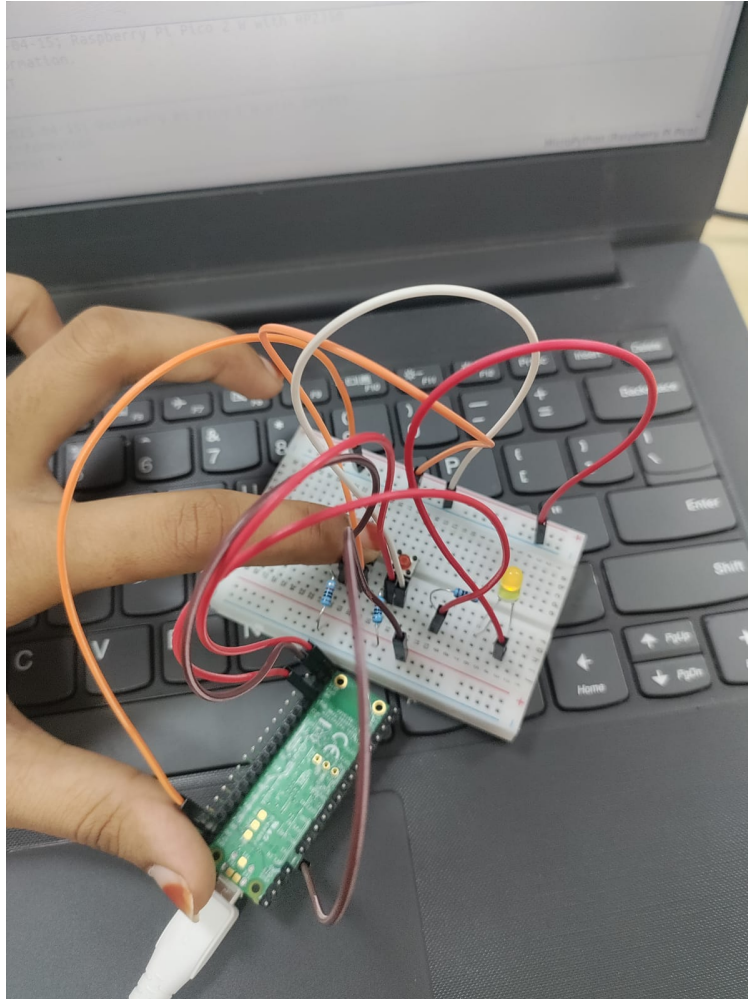
$$\text{NAND}(A, B) = \overline{A \cdot B}$$

$$\text{Output} = \begin{cases} 1 & \text{if } A = 0 \text{ or } B = 0 \\ 0 & \text{if } A = 1 \text{ and } B = 1 \end{cases}$$

4. Observation Table

A	B	LED(Output)
0	0	1
0	1	1
1	0	1
1	1	0

5. Circuit Image



6. Applications of NAND Gate

- Used to implement all other basic logic gates (AND, OR, NOT).
- Fundamental building block in digital electronics and processor design.
- Used in flip-flops, counters, and memory circuits like SRAM.
- Employed in Arithmetic Logic Units (ALUs) for logical operations.

7. Conclusion

This hardware implementation successfully verifies the theoretical behavior of a NAND gate using Raspberry Pi Pico2W and basic electronic components. NAND, being a universal gate, is a crucial part of digital design. Understanding and demonstrating it physically enhances our knowledge of logic gate operations and their importance in digital circuit design.

8. References

- GitHub Repository: https://github.com/Rithrishareddy/RITHRISHA_FWC/tree/main/Hardware
- Raspberry Pi Pico Documentation
- GATE 2009 EEE Question Paper